

Introduction to Linux/UNIX Systems

Linux User Management

Linux Users and Groups

- User and group names need to be alphanumeric strings (convention):

`[a-z_] [a-z0-9_-]* [$]?`

- Internally: Unique IDs (integers) for both users and groups
- User ***must*** be a member of one “main group” (= *Login Group*) – stored in `/etc/passwd`!
- User ***may*** be member of additional groups – stored in `/etc/group`!

Linux User Management: Commands

◆ Users

◆

◆ **useradd**

◆ **usermod**

◆ **userdel**

◆ **passwd**

◆ Groups

◆

◆ **groupadd**

◆ **groupmod**

◆ **groupdel**

Linux User Management: useradd

Usage: **useradd** *options* ... *username*

Username ... name of new user (account name) → *Example: fred*

Important options (see also: **man useradd**):

-d PathToHomeDir: Absolute path to user home directory .

→ *Example: -d /home/fred (but it will not be created by this option!)*

-s Shell: Login shell (standard command line interpreter to use)

→ *Example: -s /bin/bash*

-c Comment: User information = Full user name

→ *Example: -c "Fred Flintstone"*

-m: Create the user home directory passed with **-d** option ("make dir")

-g group: "Main group" (*Login Group*) of user

→ *Example: -g flintstones*

-G group2,group3,...: Membership in additional **groups** (*no spaces!*)

Linux User Management: Example

How to remember parameters of useradd:

Getting something done makes charts green.

das chamäleon mag gerne Grüne schmetterlinge.



Example:

```
useradd -d /home/bob -c "Bob Hawk" -m \  
-g birdfriends -G students,htl3,it -s /bin/bash bob
```

After **useradd**: set user password

- interactive -- shell: **passwd hans**

- in shell-script: **echo username:password | chpasswd**



Linux User Management: Files

1. `/etc/passwd` → *User*
2. `/etc/shadow` → *Passwords*
3. `/etc/group` → *Groups*

Linux User Management: `/etc/passwd`

- ◆ Text file `/etc/passwd`:
 - One line for each user
 - Fields separated by “:”
- ◆ Format:
 - ◆ `username:password:UID:GID:user_info:home_directory:shell`
 - ◆ `fred:x:501:501:Fred Flintstone:/home/fred:/bin/bash`
 - ◆ `testuser:x:502:502::/home/testuser:/bin/sh`
 - ◆
 - ◆
 - ◆ Attention: Login group (main group) of user (GID) is defined in `/etc/passwd`!

Linux User Management: /etc/shadow

- ◆ Text file **/etc/shadow**:

- One line for each user
- Fields separated by ":"
- Contains "encrypted" passwords (e.g. MD5 Hash)

- ◆ Format:

- ◆ **username:password:last_change:MinD:MaxD:Warn:Exp:Dis:Res**

- ◆ **fred:\$1\$5rnvr/Iq\$yk/Ccq2tQrWJxHnE0:11508:0:180:14::12783:**

- ◆
 - last_change: days since Jan 1, 1970 that password was last changed
 - MinD: days before password may be changed (0 ... anytime)
 - MaxD: days after which password must be changed (99999 ... never)
 - Warn: days before password is to expire that user is warned (7 .. default is 1 week)
 - Exp: days after password expires that account is disabled (usually empty)
 - Dis: days since Jan 1, 1970 that account is (will be) disabled (usually empty)
 - Res: a reserved field

Linux User Management: /etc/group

- ♦ Text file **/etc/group**:

- One line for each group
- Fields separated by “:”
- Members separated by “,” (*no spaces!*)

- ♦ Format:

- ♦ **groupname:password:GID:username1,username2,...**

- ♦ `accounting:x:553:gosch`

- ♦ `sytbs_3di:x:554:beer,biglmaier,bozsoki,dilch,eiweck,galk
a,gornik,hiess,holzer,huber,kadnar,lackner,lier,obentrau
tf,obentrautp,paar,paradzikovic,prinzinger,reiter,schick
,setnicka,straka,unger,vitek,wimmer,zartl,ziwohlava`

Linux User Management: More Info

(German)

(1) <http://www.fibel.org/linux/>

(2) <http://www.selflinux.org/>

(3) <http://www.netzmafia.de/skripten/unix/>